Final Documentation & Project Summary

Project: Stretto Technologies Enterprise Database System

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GitHub Link: https://github.com/EricGbagi/advanced-database-portfolio-MIT8103-Omafuvwe-Gbagi

1. Executive Summary

1.1 Background

Stretto Technologies is an e-commerce and electronics sourcing platform based in Lagos. The company specialises in sourcing high-value consumer electronics. A dynamic model was used to source items FOR sale. This involved NOT keeping a static inventory with high warehousing costs, but sourcing inventory on-demand. When a customer wishes to purchase, inventory will be sourced from trusted vendor sources.

1.2 Enterprise Objectives

A comprehensive multi-paradigm database system was developed with 5 modules to provide a robust database system. This would provide an efficient and rigorous database design to manage financial accountability. While having the flexibility to provide cloud distribution.

Within the business model, the multi-modular system has been designed to address:

i. Relational Database (PostgreSQL): Provide a database to handle financial reconciliations. This includes handling 50,000 records, multi-item invoices, ACID Transactions, and item level sourcing.
ii. Query Optimization: B-Tree Indexes (using EXPLAIN ANALYZE) were implemented along with advanced querying to improve speed when retrieving data from time critical reports.
iii. Transactions & Concurrency: To prevent concurrency issues, lost updates and phantom reads, serializable Isolation Levels were applied. Along with row level locks (SELECT ... FOR UPDATE).
iv. NoSQL (MongoDB Atlas): A document orientated schema was implemented to provide fast denormalised querying. This allows FOR highly variable product specs.
v. Cloud Distribution: A fully managed cloud was created using the native Replica Sets (High Availability using primary/secondary nodes), sharding (horizontal distribution using customer_id) and enforcing CP consistency from CAP Theorem.

2. Evidence of Work

The GitHub repository contains 5 sections to show all work completed. This includes the full technical designs, SQL schemas, query plans, and evidence of cloud executions.

a. **Portfolio 1 (portfolio-1-database-design.pdf):**

This provides the foundational design FOR the database. A fully normalised database with 6 tables following 3NF standard. An Entity Relationship Diagram (ERD) showing clear primary/foreign keys. Along with clear constraints and scripts to generate high volume test data using generate_series in PostgreSQL.

b. **Portfolio 2 (portfolio-2-query-optimization.pdf):**

This provides advanced query optimization. Three complex queries were created. These include filtering by date range, aggregating supplier revenue using joins and aggregate functions, and reconciling outstanding balances using LEFT JOIN and COALESCE. B-Tree indexes were created on high cardinality time fields. Basic query plans (using EXPLAIN ANALYZE) and an analysis after optimization showing how queries moved to Bitmap Heap Scans.

c. **Portfolio 3 (portfolio-3-transactions-concurrency.pdf):**

This provides evidence of concurrency control. ACID compliance was achieved by effectively committing transactions. Along with enforcing constraints to rollback if there were any NOT NULL issues. High level SERIALIZABLE isolation levels were implemented to ensure accurate financial auditing. Row level locking was also looked at to reduce UPDATE anomalies.

d. **Portfolio 4 (portfolio-4-nosql.pdf):**

MongoDB Atlas was used to provide a Document Store. This included a nested JSON document to represent a full Stretto Technologies transaction. Evidence of implementation including inserts and filtering queries. This also explains why the schema is good FOR variable electronics.

e. **Portfolio 5 (distributed-cloud.pdf):**

This describes distributed cloud Database Systems. The benefits of moving to the cloud, high availability using Replica Sets (ensuring replica set health using primary/secondary nodes), horizontal partitioning using sharding. CP consistency was favoured when looking at the CAP Theorem. Evidence of running on the cloud**.**

3. Reflection

The database design FOR Stretto Technologies provided me with a deeper understanding of balancing factors within modern data engineering. As no one model will suit all needs, shown by the move from a highly normalised relational database (PostgreSQL) to a non-schema document database (MongoDB Atlas).

MongoDB provided great flexibility FOR highly variable product catalogues. While PostgreSQL provided strong referential integrity, financial consistency and complex joins FOR financial

auditing. The challenges of scaling in the cloud and providing high availability were shown using Replica Sets and sharding within MongoDB Atlas. Overall, **this helped improve my** knowledge of query optimization, transaction security and cloud management. It also helped bridge the gap between designing databases and implementing them within real life enterprises.

4. References

Global Development Group FOR PostgreSQL (2026). Concurrency Control, Optimisation Tips & SQL Commands. Available from: https://www.postgresql.org/docs/16/index.html

MongoDB, Inc. (2026). MongoDB Manual. Available from: https://www.mongodb.com/docs/manual/

Elmasri, R. & Navathe, S.B. (2015). Foundations of Database Systems (7th Edition). Pearson.

Coronel, C. & Morris, S. (2018). Database Systems: Design, Implementation, & Management (13th Edition). Cengage Education.

Declaration

Generative AI (Gemini) was used throughout this project to help with formatting markdown, troubleshooting SQL issues and aiding with brainstorming ideas FOR the case study FOR Stretto Technologies. All logic FOR database design, execution, query optimization and final work was verified and completed by the author.